

Novel Test Device for Non-destructive Experimental Characterization of Void Evolution in Microscale Solder Joints subjected to Thermal Aging

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Abstract—There is a trend towards reduction in micro-bump size and pitch in 2.5D/3D packaging driven by increased bandwidth and power efficiency requirements. One of the reliability challenges that arises is the void evolution in the solder joint volume of the micro-bump accompanied by the growth of Cu-Sn IMCs in the Cu pillar after thermal aging for long periods of time. In this study, we describe a methodology to fabricate test devices with multiple sub-30 μm pitch Cu-Sn-Cu junctions, with the capability to observe the growth of voids in the micro-bumps in a non-destructive manner. Typically used material and void characterization methods for microbump interconnects are destructive, requiring cross-sectioning and polishing of the samples before observation of voids in solder joints. The cross-sectioning is cumbersome, results in loss of data, and once cross-sectioned, the samples cannot be used for further thermal aging. A non-destructive method of characterization assures continuity in the experiments during the void evolution.

The fabricated test device consists of multiple sub-30 μm pitch Cu-Sn-Cu junctions that are directly observable under a Scanning Electron Microscope (SEM). This setup can be used to do experiments ex-situ (by periodically pausing experiments, to measure void growth under SEM) or in-situ (with measurements of void growth in real-time under SEM). The test devices can also be subjected to current via a 4-wire Kelvin sensing probe setup attached to 3-axis piezo stages with a resolution of 50 nm. Multiple devices with varying sizes of solder bumps are fabricated on a single die. The dies are aged in an environmental chamber at a temperature of 175 °C for hundreds of hours. They were periodically taken out of the environmental chamber and the void evolution in the solder joint on the devices measured under both optical microscopes and under an SEM. Finally, the effect of different bump sizes on the reliability of the solder joints is reported in this study.

Index Terms—Copper Pillar Solder Microbumps, Solder Interconnects, Non-destructive Characterization, Thermal Aging, Solder Voiding

I. INTRODUCTION

Requirements for higher bandwidth, processing speeds and better power efficiency are driving an industry trend towards reduction of the size and pitch of Solder microbump interconnects used in heterogeneously integrated packages. Such

ultra-fine pitch Cu pillar Solder microbumps pose significant reliability challenges due to electromigration and/or thermomigration induced voiding in the solder joint, accompanied by the growth of Cu_6Sn_5 intermetallic compound (IMC) in the Cu pillar. This challenge is particularly significant in Cu pillar-SnAg microbumps of diameters under 100 μm . The scanning electron micrograph in Fig. 1 illustrates the failure phenomenon under thermal aging in a microbump. Similar observations of the formation of Cu-Sn IMCs on the Cu pillar sidewall under simultaneous current and temperature stressing resulting in voiding in the solder joint volume was reported in [1]. This phenomenon was observed in solder bump of 20 μm diameter on being subjected to thermal aging at a temperature of 150 °C and current density of the order of 10^4 A/cm^2 (see Fig. 2).

The typical procedure for characterizing Cu-Sn intermetallic growth and voiding in such solder joints involves cross-sectioning and polishing of the microbump after it has been subjected to current and/or thermal stressing. This characterization procedure is destructive in the sense that the devices cannot be used for further experimentation. Destructive evaluation prevents continuous observation of phase evolution in the microbumps.

The design of the test device for non-destructive characterization of microscale solder samples is inspired by the Blech-like device fabricated for observing electromigration void evolution in Cu metal lines [3]. The test devices for non-destructive characterization of Cu-Sn intermetallic growth and solder joint voiding designed in this paper have the capability to be observed under the Scanning Electron Microscope (SEM) in real time (*in-situ*) or can be periodically taken to the SEM for observation after being subjected to the current/thermal stressing for a period of time (*ex-situ*).

II. TEST DEVICE DESIGN

The test structure design for the non-destructive characterization of Cu-Sn IMC growth and voiding in the ultra-fine pitch Cu pillar solder microbumps is shown in Fig. 3. The device has four contact pads, two each for probes corresponding to the

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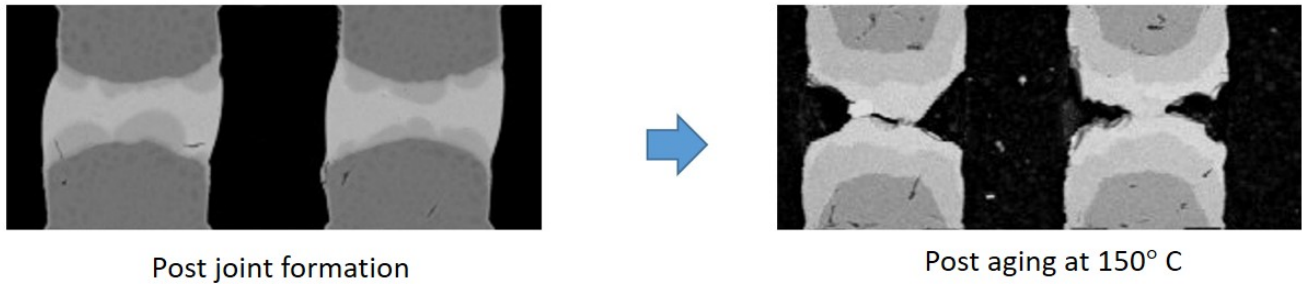


Fig. 1: SEM cross-section of Cu-SnAg microbump.

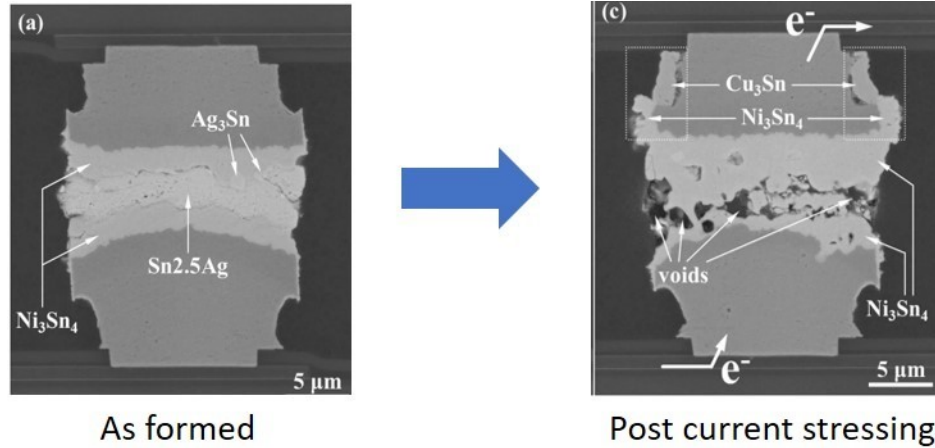


Fig. 2: Electromigration induced Cu-Sn IMC growth on Cu pillar sidewall on current stressing [1].

voltage and current terminals for Kelvin 4-wire measurement. Multiple segments of electroplated tin (Sn) fill the gaps in the copper lines allowing multiple observations of voiding in the Sn channels simultaneously. Each die has 20 test structures with varying lengths of the Cu and Sn segments in different structures.

The test device replicates the geometrical dimensions and the material interfaces of a typical Cu-solder microbump assembly as shown in Fig. 4. Therefore, it enables observation of Cu-Sn IMC growth and voiding in solder channels under combined current and thermal stressing, when the devices are subjected to electromigration at elevated temperatures. The evolution of the Cu-Sn IMCs and voids can be observed without the typical destructive process of cross-sectioning and polishing [4], [5]. This device design is capable of maintaining continuity in experiments, that is, the samples can be subjected to current/thermal stressing for a period of time, then examined under an SEM, and then subjected to further current/thermal stressing.

This device design is intended to be used as a template test vehicle by allowing different Solder compositions such as SAC 305 or SnBi for example in place of pure Sn used in Solder channels in this study. After experimentation on such samples with the various Solder compositions, the growth of Cu-Sn IMCs at the Cu-Solder junctions and the voiding in the Solder volume can be observed and their rates can

be estimated. Therefore, this test device and experimental methodology can be used to compare the reliability of various Solder compositions when subjected to thermal aging and electromigration at elevated temperatures.

The base layer chosen for this study consists Titanium (20 nm), Copper (500 nm), Nickel/Cobalt (300 nm) and Gold/Palladium (10 nm). Titanium serves as an adhesion layer as Cu has poor adhesion to the silicon wafer. Nickel/Cobalt is selected as a diffusion barrier layer to prevent Cu in the base layer from diffusing into the solder from that interface. A thin layer of Palladium/Gold serves as a passivation layer to prevent oxidation of the base layer. Ideally, electromigration driven voiding would occur simultaneously in all the solder channels along the test structure. In practice, nucleation and void growth is likely to occur at different instants for different Cu-solder junctions on the same device. Assuming a scenario where one of the Cu/solder junctions develops a stable void, the design allows the current to follow an alternate path through the base layer. A typical thin base layer will have higher resistance and significant Joule heating. To mitigate joule heating a Cu base layer of 500 nm thickness is used for the test device.

III. FABRICATION OF TEST DEVICE

In this section, the steps involved in the fabrication of the designed test devices for non-destructive characterization of Cu pillar solder microbumps is discussed. The overall

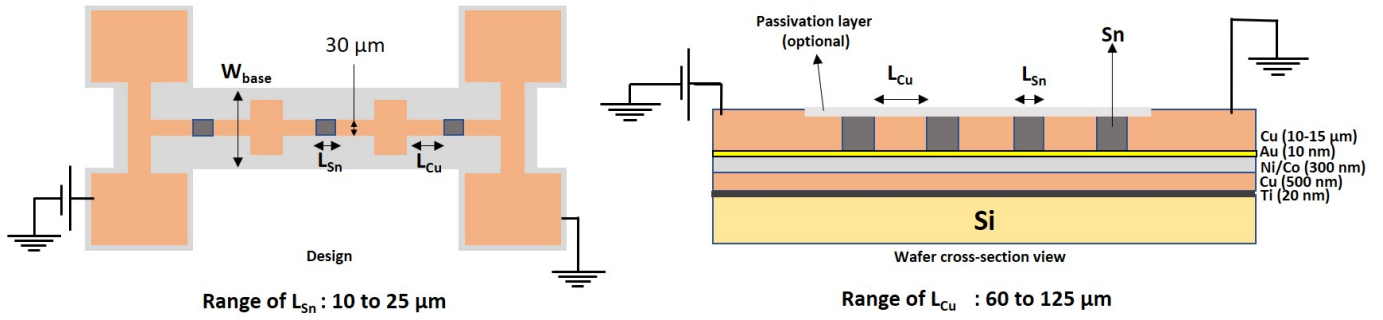


Fig. 3: Test Device Design Schematic.

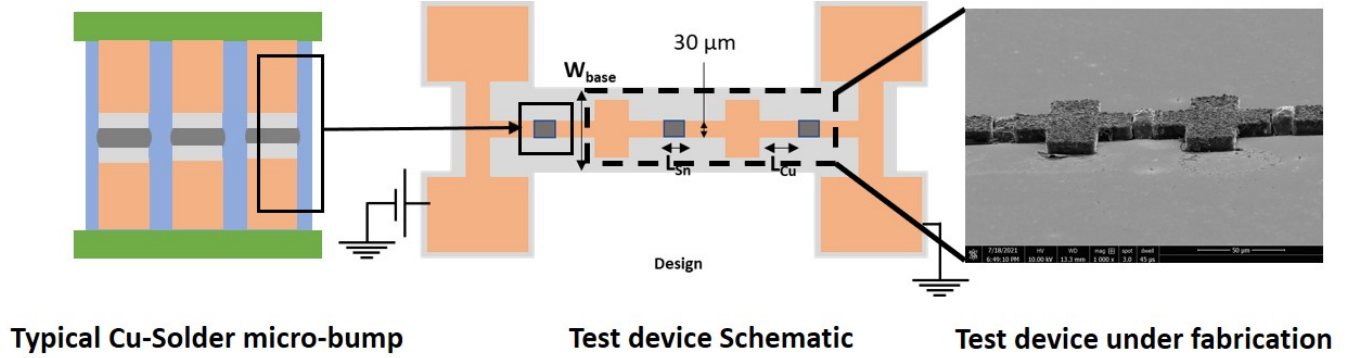


Fig. 4: Schematic showing how designed test structure corresponds to actual microbump.

fabrication process flow is shown in Fig. 5. The fabrication process was carried out in the Scifres cleanroom facilities at Birck Nanotechnology Center at Purdue University.

The different materials which form the base layer were deposited without breaking vacuum using evaporation sputtering or physical vapor deposition to avoid oxidation. Since the electroplated structures are 15 μm to 30 μm in height, we used a recipe with double spin coat of AZ 9260 photoresist (PR). The patterning of the thick PR was done via non-contact laser aligners for photolithography. The copper and solder structures were electroplated on the base layer. Controlling the SnAg composition during the electroplating was a challenge and therefore for these sets of devices, pure Sn was plated instead. A small amount of Ag improves thermal fatigue resistance and mechanical strength. Since our experimental goal is to study electromigration and thermomigration, using pure Sn is not expected to affect the results.

The test devices underwent solder reflow with a peak temperature of 270 $^{\circ}\text{C}$. The reflow was performed under Ar gas environment using a heating stage as shown in Fig. 6 to prevent oxidation of Cu and Sn at high temperatures. The SEM images of the Sn channel before and after reflow are shown in Fig. 7.

For tuning the resistance of the device for electromigration experiments, the base layer was etched to different dimensions depending on the target current density in the Sn channel for electromigration experiments. The resistance of each device was measured using the 4-probe electromigration test setup

shown in Fig. 9. The analytical sheet resistance of the channels were calculated using the sheet resistance of the various materials on the device, assuming that the base layer and the Cu-Sn channels are current paths parallel to each other. The equivalent circuit representation of the device with the base layer and Cu-Sn channels is shown in Fig. 8. The base layer was wet-etched after photoresist (PR) patterning of the wafer after solder reflow, corresponding to the dimensions of the parameter W_{base} using dilute Aqua Regia ($\text{HCl} : \text{HNO}_3 : \text{H}_2\text{O} = 3 : 1 : 1$). Additionally, in case of voiding in the solder channel, the base layer provides an alternate pathway for the current in our test structure.

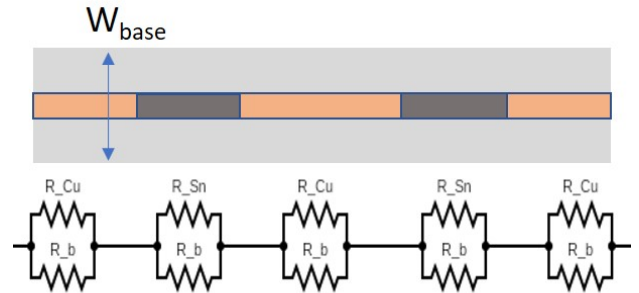


Fig. 8: Circuit representation of the test structure undergoing electromigration.

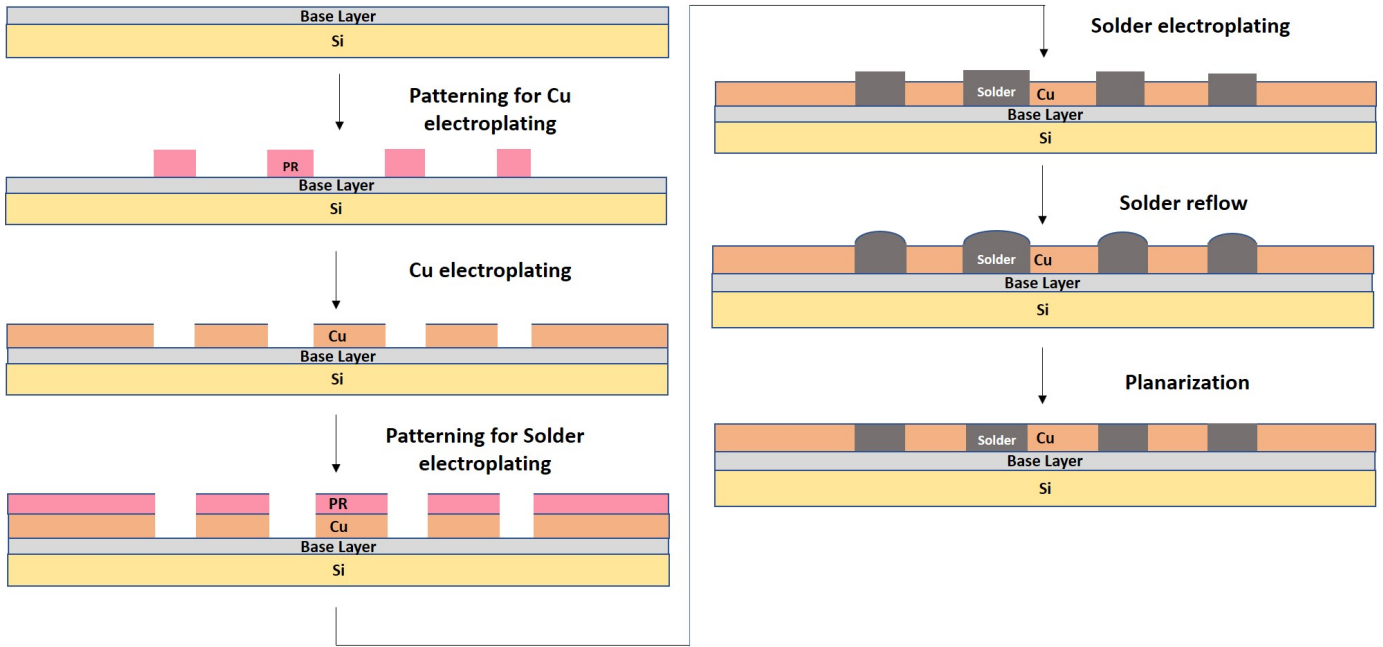


Fig. 5: Fabrication Process Flow Chart.

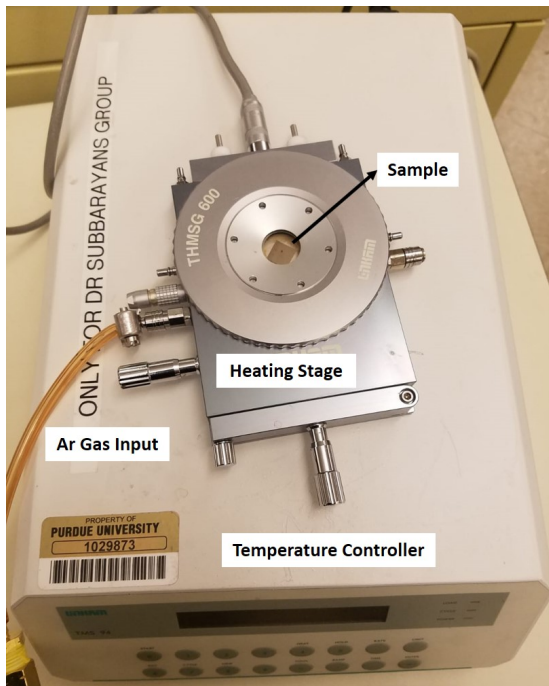


Fig. 6: LINKAM Heating Stage Setup for Solder Reflow.

IV. EXPERIMENTAL SETUP AND PLAN

A. Thermal Aging

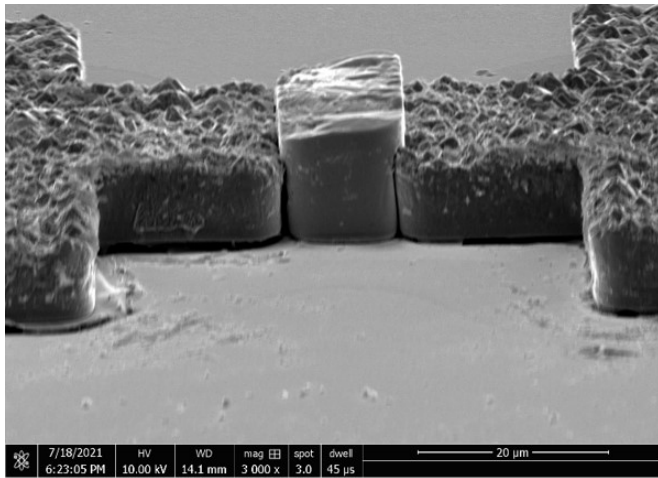
The thermal aging of samples without current stressing were performed in environmental test chambers by aging the test samples at a constant temperature for a very long period of time, over 1000h. Significant voiding in solder microbumps

was reported at temperatures of 150 °C over aging times of approximately 1000 hours (see Fig. 1). Optical microscope was used for preliminary visual checks for solder voiding and growth of intermetallics on the Cu line. The optical images were not accurate enough or reliable enough to measure voiding or intermetallic growth. The voiding in the solder joint was then observed by taking the device out of the environmental chamber at different time intervals and then observing the Cu-Sn-Cu junctions using an SEM. A trench was also milled using Focused Ion Beam(FIB) along the Cu-Sn junction in the device after aging. The various material phases, including the Cu_6Sn_5 phase at the Cu-Sn junction were then identified along the wall of the FIB trench using elemental analysis done using Energy-Dispersive X-ray spectroscopy(EDX).

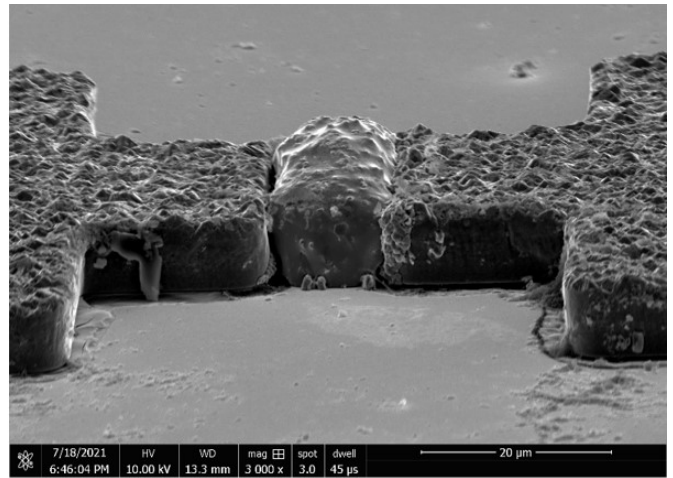
Solder joint voiding under thermal aging is a significantly slower process than voiding under electromigration at high current densities [6]. Therefore, the electromigration experiments at elevated temperatures and thermal aging experiments need to be performed separately in different environments and using different experimental test setups.

B. Electromigration

The experimental setup has 4 probes that are driven via piezo-activated stages capable of a six-axis motion with a position resolution of 50 nm. The test setup is designed to fit into the chamber of an SEM. The dimension and weight of the test setup are mainly driven by this consideration. The samples may be heated to elevated temperatures through a localized resistive ceramic heater placed under the sample. The experimental setup is shown in Fig. 9, and the image on the right illustrates the ability to fit inside an SEM.



Before Reflow



After Reflow

Fig. 7: SEM image of the Sn channel before and after reflow.

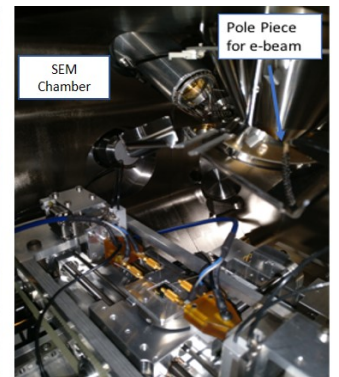
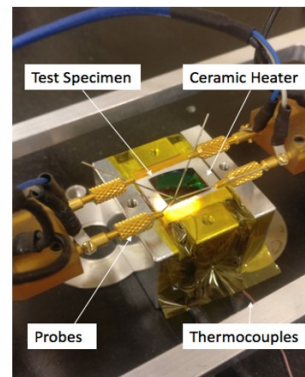
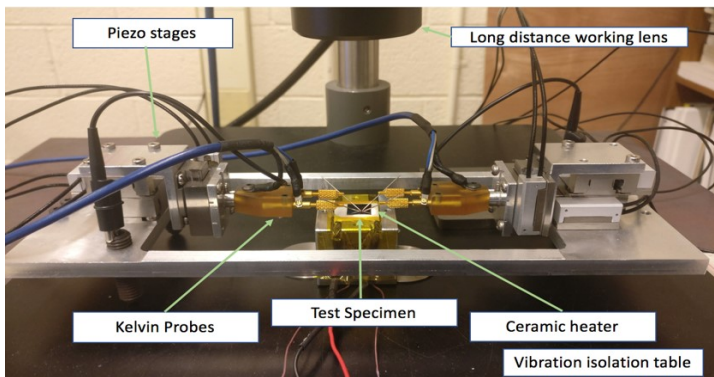


Fig. 9: Nano-resolution 4-probe experimental setup.

The typical procedure followed for electromigration testing of the samples is shown in Fig. 10. The samples are to be tested for a duration of time and then imaged under the SEM to characterize the Cu-Sn intermetallic evolution and voiding, if any. An SEM that is capable of EDS can identify the different Cu-Sn intermetallics such as Cu_3Sn and Cu_6Sn_5 . Prior literature report effects of electromigration in solder at current densities in the range of $10^4 - 10^5 \text{ A/cm}^2$ for solder in flip chip and microbump interconnects [5], [7], [8]. Consistent with the literature, the present test device dimensions will subject the sample to current densities of the order of $10^4 - 10^5 \text{ A/cm}^2$.

V. RESULTS

A. Thermal Aging

The SEM images of results of the thermal aging of the test structures having solder channels of lengths $15 \mu\text{m}$ and $30 \mu\text{m}$ at 175°C at 600h, are compared with the SEM images of the channels after reflow in Fig. 11. The profile of the voiding in the solder bumps are similar to the solder joint voiding observed in Fig. 1.

The voiding in the solder channel in samples aged at 175°C for 0, 50 and 200 h are shown in Fig. 12. The SEM images were obtained at different thermal aging times by imaging the same solder channels after various aging times. As the aging time increases, the depletion of Sn from the solder channel volume resulting in voiding in the channel can be observed. The depletion of Sn was accompanied by the growth of Cu_6Sn_5 at the Cu-Sn junction. The imaging of the Sn channel at the same location on the die, after various aging times demonstrate the ability of the test structure design to be used for further experiments after each thermal aging step.

Voiding was observed to stop completely in the solder joint volume after thermal aging for sufficient time. Preliminary observations on the voiding rate for the $15 \mu\text{m}$ and $30 \mu\text{m}$ wide Sn channels suggest that the void growth stops earlier in the smaller sized channel as compared to the larger sized channel. This might be due to less Sn being available in the solder volume for formation of Cu-Sn intermetallics. Extensive imaging of the solder joint volume or different channel sizes for different aging times is required to accurately characterize the voiding rate quantitatively.

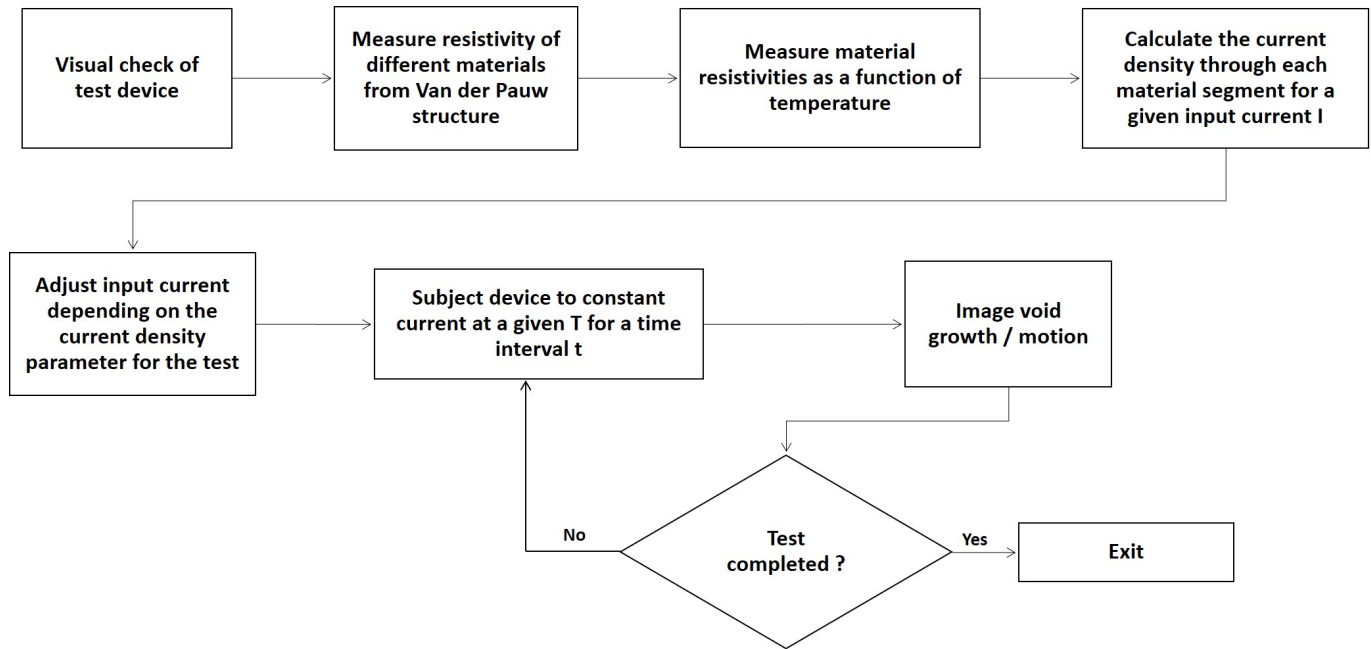


Fig. 10: Flowchart of the experimental test procedure for electromigration.

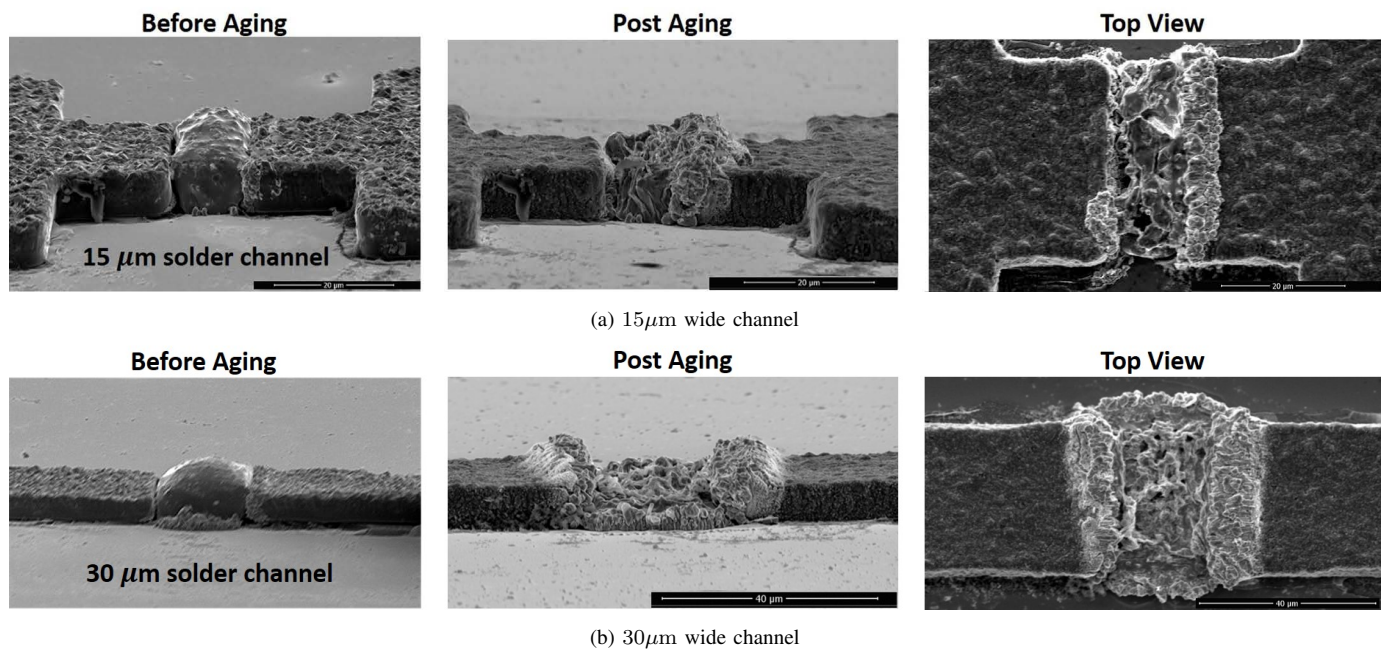


Fig. 11: Thermal aging of test devices at 175 °C for 600 h.

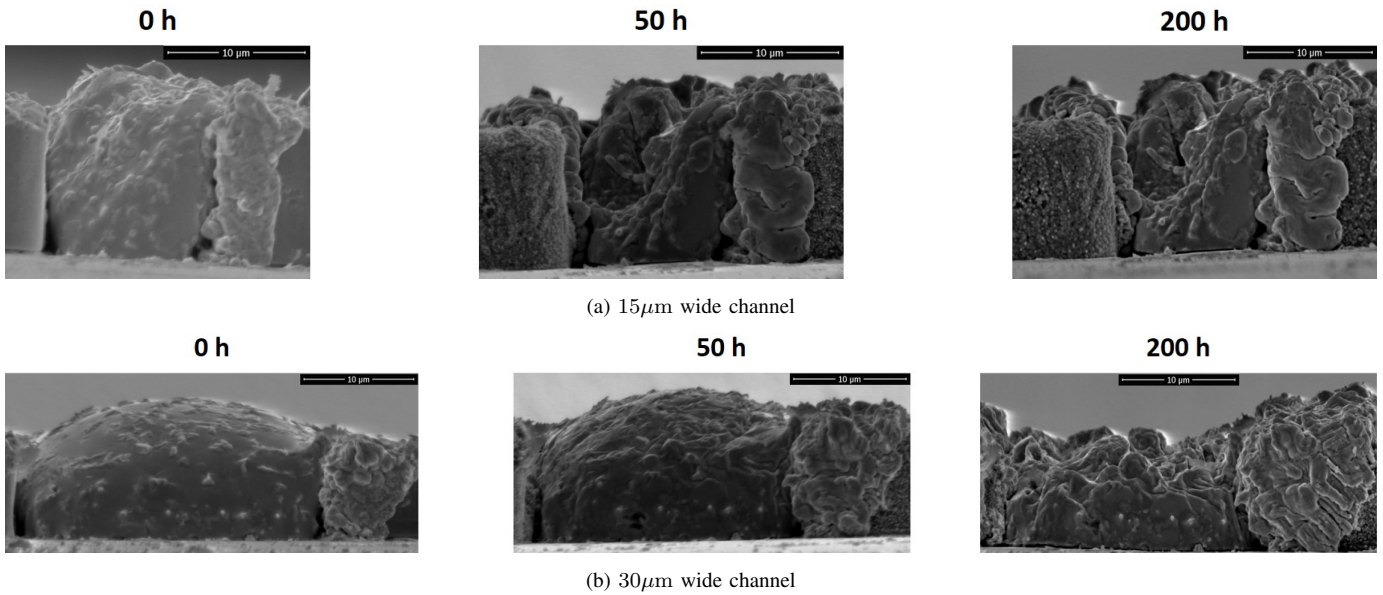


Fig. 12: Voiding in solder channels thermally aged at 175 °C over 200 h.

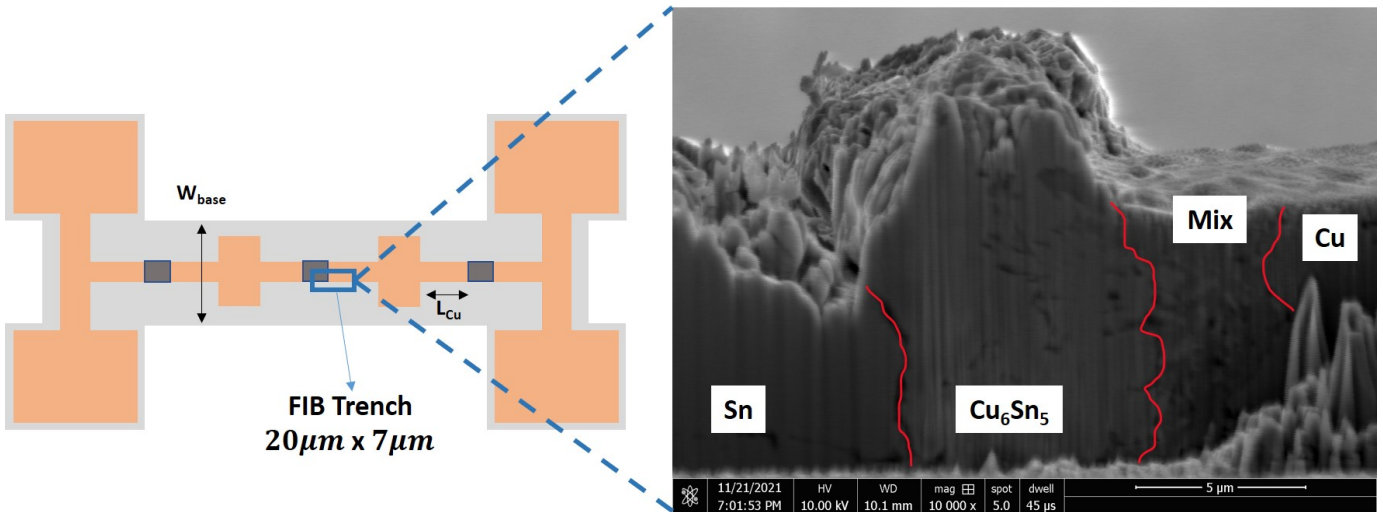


Fig. 13: FIB cut at Cu-Sn junction showing the different material phases after Thermal aging at 175 °C for over 1000 h.

The 2D SEM images of the microbumps were found to be not reliable enough to provide accurate quantitative estimation of volume of voiding in the solder joint for different aging times. Imaging of the voided microbumps using 3D X-ray Microscopy is required to accurately quantify void growth in the Solder volume for different aging times. Test devices with different Solder channel dimensions will be thermally aged for different aging periods at different temperatures, and imaged using 3D X-ray microscopy to get an accurate quantitative estimation of the rate of voiding in the solder volume in future studies.

To characterize the growth of Cu-Sn intermetallics such as Cu₆Sn₅ in the Cu pillar solder microbump assembly, Focused Ion Beam (FIB) was used to mill a trench cutting into the test structure at the Cu-Sn junction between the Cu and Sn

channels. The dimensions of the trench and the location on the test structure are shown in Fig. 13. The FIB milling of the test structure results in a smooth finish similar to cross-sectioning and polishing, which can be used to perform Energy-Dispersive X-ray spectroscopy (EDX) to characterize the various material phases at the Cu-Sn junction. The various material phases were identified using EDX and labelled, as shown in Fig. 13. Apart from the Sn, Cu and the Cu₆Sn₅ phase present at the Cu-Sn junction, there is also a mixed phase with both Cu₆Sn₅ phase and another phase which could not be accurately identified with EDX.

Further experiments to characterize solder channel voiding and Cu₆Sn₅ growth via FIB milling for samples aged at lower and higher temperatures than 175 °C are planned for the future. Characterization of intermetallic growth in FIB milled

cross-sections on samples with different Sn channel widths could provide more insight on the preliminary observations on differences in voiding rates with Sn channel width.

B. Electromigration

Literature indicates electromigration induced Cu-Sn intermetallic growth and voiding causing a resistance increase of about 50% total in microbumps of 30 μm diameter [9] for time period not specified. In other literature pertaining to electromigration induced Cu-Sn intermetallic growth, voiding was observed at the cathode for electromigration tests for time periods in the range of 80 h at 180 °C at a current density of $3.2 \times 10^4 \text{ A/cm}^2$ in the solder channel [6].

Electromigration experiments were performed at 150 °C for a current density of $2.4 \times 10^4 \text{ A/cm}^2$ through the solder channel for 75 h. These experiments did not show any substantial increase ($\leq 2\%$) in the resistance, indicating that there was no significant voiding in the solder channel induced due to electromigration in that time period. Therefore, the next set of electromigration experiments were performed at another test structure at 175 °C for a current density of $4.1 \times 10^4 \text{ A/cm}^2$ for 25 h. This experiment showed around 10% increase in the resistance of the device, but did not show any discernible voiding in the solder channel volume when viewed under the SEM. More electromigration experiments are planned at higher current densities for longer periods of time in combination other characterization techniques such as 3D X-ray Microscopy in the future.

VI. SUMMARY

A novel test device capable of non-destructive characterization of Cu-Sn IMC growth and voiding under thermal and/or current stressing was designed and fabricated. The fabrication process flow was described in detail. Some results from thermal aging and electromigration experiments on the test structures were discussed and the non-destructive nature of the experimentation and characterization procedure of Cu-Sn intermetallics in Cu-Sn microbumps were demonstrated.

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REFERENCES

- [1] Y.-W. Chang, C.-c. Hu, H.-Y. Peng, Y.-C. Liang, C. Chen, T.-c. Chang, C.-J. Zhan, and J.-Y. Juang, "A new failure mechanism of electromigration by surface diffusion of sn on ni and cu metallization in microbumps," *Scientific reports*, vol. 8, no. 1, pp. 1–10, 2018.
- [2] S. Prasad, C. Jois, and G. Subbarayan, "Device and test designs for in-situ observations of phase evolution in solder micro-bumps due to electromigration and thermal aging," in *TECHCON*, 2021.
- [3] Y. Singh, "Characterization of interfacial energy of thin films through current induced diffusive interfacial voiding," Ph.D. dissertation, Purdue University Graduate School, 2019.
- [4] S. Ren, M. Sun, Z. Jin, Y. Guo, H. Ling, A.-m. Hu, and M. Li, "Formation mechanism of novel sidewall intermetallic compounds in micron level sn/ni/cu bumps," *Electronic Materials Letters*, vol. 15, no. 5, pp. 562–571, 2019.
- [5] J.-A. Lin, C.-K. Lin, C.-M. Liu, Y.-S. Huang, C. Chen, D. T. Chu, and K.-N. Tu, "Formation mechanism of porous cu3sn intermetallic compounds by high current stressing at high temperatures in low-bump-height solder joints," *Crystals*, vol. 6, no. 1, p. 12, 2016.
- [6] K.-N. Tu, *Solder joint technology*. Springer, 2007.
- [7] T. Lee, K.-N. Tu, S. Kuo, and D. Frear, "Electromigration of eutectic snpb solder interconnects for flip chip technology," *Journal of Applied Physics*, vol. 89, no. 6, pp. 3189–3194, 2001.
- [8] T. Lee, K.-N. Tu, and D. Frear, "Electromigration of eutectic snpb and snag 3.8 cu 0.7 flip chip solder bumps and under-bump metallization," *Journal of Applied Physics*, vol. 90, no. 9, pp. 4502–4508, 2001.
- [9] T.-W. Lin, K.-C. Shie, C. Chen, and K.-N. Tu, "Study of electromigration of solder microbumps by 3d x-ray microscopy," in *2020 15th International Microsystems, Packaging, Assembly and Circuits Technology Conference (IMPACT)*, 2020, pp. 76–78.